Jumping Into the Frame

Reconstructing Interactive, Physically-Grounded 3D Environments from a Single Image

Joshua Ford
Advisor: Marie-Paule Cani

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Abstract

Single-image 3D scene generation has advanced rapidly, but most systems stop at a viewable, inert result: a panorama, a layered mesh, or a radiance field that supports novel-view synthesis but has no motion and nothing to interact with. We present **Into Frame**, a system that turns one photograph into an *inhabitable* environment, in which discrete objects exist as individual simulated assets, ground cover and vegetation respond to contact and wind, and the scene is lit by a physically based estimate of the original illumination, all at real-time VR frame rates. Our central design commitment is to reconstruct each part of the scene by a method suited to its physical role, rather than extracting everything from one representation: terrain, objects, ground cover, water and sky are handled separately and by different means. We outline the pipeline built on this principle and focus on four contributions: a projection from calibrated panoramic depth to a top-down height field that preserves each cell's observed panorama pixel; a terrain reconstruction that keeps visual similarity with the original image while applying fluvial erosion to give plausible relief where nothing was observed; a pattern-texturing scheme that avoids the projective smearing an equirectangular photograph produces on the ground; and a point-process population synthesis that scatters vegetation and objects to match the spatial statistics of the real detections. We evaluate the system on several real-world images and report both what it does well and a set of failure modes we were able to measure precisely. We argue that the most transferable result is methodological: at this scale, most defects are not model failures but *assumption* failures, where a step valid for one scene category is silently applied to another.

1 Introduction

1.1 Motivation

A photograph is a two-dimensional projection of a three-dimensional world, and recovering that world from a single view is underconstrained: infinitely many scenes project to the same image. Humans nonetheless read a photograph as a place. We infer that the ground continues past the frame, that the occluded side of a rock exists, that the flowers in the foreground are individual plants rather than a printed texture. Building an environment that supports those inferences, one a person can *jump into*, is the problem this project begins to address.

Photographs are among the most abundant records of place we have: personal archives, cultural heritage documentation, scientific field imagery, and now, with generative models, images of places that never existed at all. A system that converts such an image into an environment a person can walk into changes what those archives are able to express. It also lowers the cost of 3D content authoring by several orders of magnitude relative to manual modeling or multi-view capture, and because the process is automatic, it puts that authoring within reach of a non-expert.

1.2 The Challenge

The difficulty is not one hard sub-problem but the interaction of several, each of which degrades the others:

Extreme ill-posedness.
A single view fixes neither absolute scale nor the geometry of anything it does not see. Roughly 5/6 of a full 360° panorama generated from a standard photograph is synthesized rather than observed, and everything behind every foreground object is unknown. Much of the world must be inferred.

Compounding error.
A pipeline that estimates depth, then geometry, then materials, then objects inherits every earlier error, and a discrepancy that is barely visible at one stage can be structural by the next.

Heterogeneous content.
The representation that suits a mountainside (a height field) is wrong for a tree (a mesh instance), which is wrong for a meadow (a scattered population), which is wrong for the sea (an animated surface). These representations must nonetheless coexist in one scene: they need to share a coordinate frame and a scale, agree on how they are lit, and not seam visibly where they meet.

Interactivity constraints.
The output must render stereoscopically at headset frame rates. That budget constrains every rendering choice, and it binds hardest on the animated, simulated elements that make the scene interactive in the first place.

1.3 Goals

Our goal is a system that takes one image and produces an environment which is *interactive* (objects animate and interact with their surroundings), *decomposed* (discrete objects exist as individual entities, not painted texture), *plausible* beyond the observed region, and *performant* at VR frame rates.

2 Related Work

2.1 Single-Image Panorama Generation

Extending a narrow-field photograph to a full 360° environment is the first step in most single-image scene systems. Diffusion models adapted to panoramic geometry are the dominant approach. CubeDiff [18] repurposes a perspective latent diffusion model for panorama generation by operating on cubemap faces with synchronized attention. DreamCube [12, 11] extends this to RGB-D panoramas through multi-plane synchronization, producing geometry alongside appearance. PanoDreamer [30] takes an optimization-based route to coherent single-image 360° scenes, and LayerPano3D [42] decomposes the panorama into depth layers to support larger viewpoint changes. Our system supports several such backends, defaulting to a FLUX-based [5] outpainting path, with WorldGen [41] and DreamCube as alternatives. The relevant distinction is that we treat the panorama as an intermediate representation rather than as the deliverable.

2.2 Single-Image 3D Scene Generation

A growing body of work targets navigable 3D scenes from one image. SphericalDreamer [34] generates navigable immersive worlds; SonoWorld [17] extends single-image scene generation to audio-visual output; 360Anything [37] lifts images and videos to 360° geometry-free representations; and WorldGen [41] produces complete 3D scenes rapidly. These systems generally optimize for visual fidelity from viewpoints near the original camera. Into Frame instead prioritizes *physical decomposition*: the output is not one geometry proxy but a set of typed entities with distinct behavior, a collidable terrain mesh, individually placed object meshes, an instanced ground-cover population and an animated water surface. Each is required before a scene can be dynamic rather than merely viewable.

2.3 Depth Estimation and Panoramic Depth

Monocular depth estimation underpins all geometry in the pipeline. We use Depth Anything 3 [24] for perspective depth and Depth Any Panoramas [25] for equirectangular depth, the latter avoiding the distortion artifacts that arise when a perspective model is applied to equirectangular imagery. A recurring difficulty, which we address explicitly in section 4, is that monocular depth is relative and saturates: real distant terrain and sky receive indistinguishable predictions, so calibrated depth at a ridge crest is an extrapolation rather than a measurement.

2.4 Terrain Modeling and Heightmap Synthesis

Recovering terrain from imagery connects to a long line of graphics work. Pixels2Peaks [16] converts terrain images to heightmaps directly; terrain modeling from feature primitives [6] builds terrain from sparse user-specified features; Earthbender [1] offers interactive stylistic heightmap generation; and WorldBrush [4] synthesizes procedural worlds from examples, including the distribution of scene elements. For physical plausibility of unobserved relief we use Landlab [2, 8, 15], an Earth-surface dynamics toolkit, applying fluvial incision so that invented terrain obeys drainage structure rather than reading as smooth interpolation or undirected noise.

2.5 Texture Synthesis

Projecting a panorama onto ground fails at grazing incidence: a single panorama row covers meters of ground at distance. Pattern-based texturing [28] addresses exactly this class of problem by assembling surfaces from patterns rather than resampling a projection, and our terrain texturing follows that idea using patches extracted from the real photograph. Where synthetic material is unavoidable we use FLUX [5] and latent diffusion models [33], with Swin2SR [3] for super-resolution.

2.6 Segmentation, Detection and Recognition

Object decomposition uses SAM 2 [32] for segmentation and video tracking, Grounding DINO [26] for open-vocabulary detection and label verification, CLIP [31] for embedding-based classification, BLIP [20] and Florence-2 [39] for captioning, Recognize Anything [44, 10] and its open-set successor RAM++ [9] for scene tagging, SegFormer [40] for semantic region typing, and BiRefNet [46] for high-resolution matting. Each solves a similar but subtly different problem; we combine them rather than choosing between them, using their agreement where they overlap to build a more reliable reading of the scene than any one provides alone.

2.7 Object Reconstruction and Inpainting

Individual objects are reconstructed with SAM 3D [36], with SPAR3D [13] and TRELLIS [38] as alternatives. Removing foreground objects so the ground behind them can be reconstructed uses inpainting: LaMa [35], MAT [21], Inpaint Anything [43], and ObjectClear [45], the last targeting removal of both objects and their effects.

2.8 Lighting

Relighting reconstructed content so that it matches the original photograph requires recovering the scene illumination. We use LuxDiT [22] for HDR lighting estimation and IntrinsicDiffusion [27] to separate albedo from shading, so that assets carry material color rather than baked-in sunlight and can be lit by our own lighting algorithm.

3 Overview

3.1 Design Principle: Role-Specialized Decomposition

The organizing commitment of Into Frame is that a scene is not one thing. Ground, discrete objects, ground cover, water and sky differ in how they must be reconstructed, how they are represented, how they are textured, and how they behave at runtime. Forcing them through a single representation means either accepting the limitations of that representation or developing a more general one, and the latter was outside the scope of this project.

Figure 1 shows the resulting structure. The pipeline begins by building three panoramic representations of the scene, each of which serves a different consumer downstream:

Original Panorama.
A synthetic equirectangular panorama generated from the input photograph by WorldGen [41], with FLUX [5] outpainting, CubeDiff [18] and DreamCube [12] available as alternative backends. This is the baseline scene representation, and the one every semantic stage reads from, because it is the only variant in which the real objects are still present.

Object-Removed Panorama.
The original panorama processed with ObjectClear [45] to remove foreground objects, followed by a FLUX LoRA correction pass that restores equirectangular consistency at the seams. This gives an unobstructed view of the environment, better suited to terrain reconstruction, where occluders would otherwise punch holes in the ground.

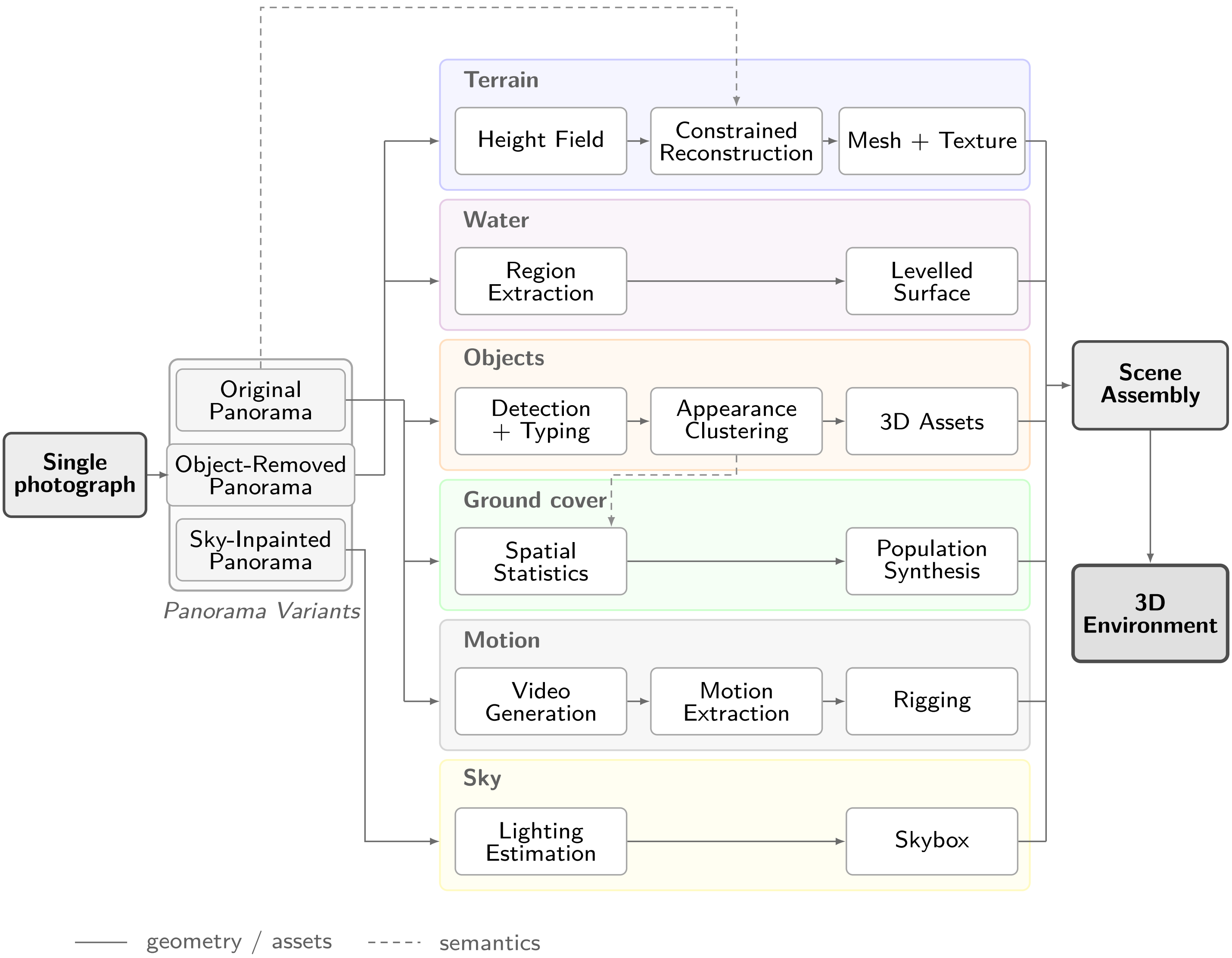


Figure 1: The pipeline decomposes a single image into six tracks with different physical roles. Solid arrows carry geometry, dashed arrows carry semantics. The three panorama variants at the left are maintained deliberately: geometry is derived from the object-removed panorama, semantics from the original.

Sky-Inpainted Panorama.
The region above the horizon, completed with FLUX [5] inpainting where objects were removed, producing a seamless sky used as both skybox and image-based lighting source.

These three panoramas are then taken in concert to generate the rest of the scene:

Terrain.
A single continuous height field, meshed with variable density, carrying collision. Reconstructed from calibrated panoramic depth and completed by a terrain constraint solver.

Water.
Regions typed as water, extracted as a separate leveled surface with an animated shader.

Objects.
Discrete entities detected, classified, clustered by visual similarity, and reconstructed across the scene.

Ground cover.
Populations too numerous to place individually as reconstructions, scattered by density over a mask derived from the semantics and color of the panorama.

Motion.
Per-object animation, guided by a generated video of the scene and used to decide which categories are rigid bodies and which need rigging.

Sky.
The panorama above the horizon, used as both skybox and image-based lighting source.

3.2 Stage Sequence

The pipeline defines 42 stages, of which 38 are active in the default configuration. Table 1 groups them by track and table 2 gives the model behind each stage that uses one. Stage numbers follow configuration order throughout. The gap at 36–37 in table 1 is two stages that are defined but disabled by default, and so belong to no track.

The panorama stages build a 360° representation; semantics identifies the objects in it; cleanup produces a second panorama with those objects removed, from which the geometry stages recover depth and metric scale. Terrain is meshed and eroded, objects are detected and their spatial distributions learned, and the scene is assembled from those statistics. Finally a video generation model guides per-object animation and decides which objects are rigid bodies.

Table 1: Pipeline stages grouped by reconstruction track.

Track	**Stages**	**Purpose**
Panorama	1–4	Caption, Perspective Depth, 360° Outpainting, Super-Resolution
Semantics	5–8	Panoramic Depth, Object Segmentation, Classification, Typing
Cleanup	9–11	Foreground Removal, Equirectangular LoRA Correction
Regions	12–15	Semantic Region Typing (original and object-removed), Lighting Estimation, Sky Inpainting
Geometry	16–19	Panoramic Depth, Metric Calibration, Height Field, Region Map
Terrain	20–25	Constrained Reconstruction, Erosion Refinement, Meshing, Texturing
Objects	26–30	Recognition, Detection, Instance Refinement, Correlation, Clustering
Population	31–33	Spatial Statistics, Population Synthesis, Ground Cover
Assembly	34–35	Asset Generation, Scene Assembly and Placement
Animation	38–42	Video Generation, Motion Extraction, Rigging, Animation

Table 2: Model behind each pipeline stage that depends on one, in configuration order. Stages that use no model are omitted, so the numbering is not contiguous; table 1 gives the full sequence. When a stage supports several backends, the default is listed first. Stages marked † are defined but disabled in the default configuration.

#	Stage	Model
1	Captioning	BLIP [20]; or Florence-2 [39]
2	Depth Generation	Depth Anything 3 [24]
3	Panorama	WorldGen [41]; or FLUX [5], CubeDiff [18], DreamCube [12]
4	Panorama Supersampling	Swin2SR [3]
5	Panorama Object Depth	Depth Any Panoramas [25]
6	Object Segmentation	SAM 2 [32], with BiRefNet [46] matting
7	Object Classification	BLIP [20] captioning with keyword matching
8	Object Typing	CLIP [31], with Grounding DINO [26] label verification
9	Panorama Inpainting†	FLUX [5]; or LaMa [35]
10	Panorama Foreground Inpainting	ObjectClear [45]
11	Panorama LoRA Correction	FLUX [5] with an equirectangular LoRA
12	Panorama Regions	SegFormer [40]
13	Panorama Regions (Terrain)	SegFormer [40]
14	Panorama Lighting	LuxDiT [22], with IntrinsicDiffusion [27] albedo/shading separation
15	Skybox Inpainting	FLUX [5]
16	Panorama Depth	Depth Any Panoramas [25]
22	Terrain Noise Refinement	Landlab [2, 8, 15] fluvial incision
25	Terrain Texture Generation	Pattern-based texturing [28], with LaMa [35] completion; or FLUX [5] with Florence-2 [39] material captioning
26	Object Recognition	RAM++ [9]
27	Object Detection	Grounding DINO [26]
28	Object Instance Refinement	SAM 2 [32]
29	Object Correlation	SAM 2 [32]
30	Object Category Clustering	DINOv2 [29], with CLIP [31] corroboration
33	Grass Cover	SAM 3D [36] for the tuft mesh
34	Panorama Asset Generation	SAM 3D [36]; or SPAR3D [13], TRELLIS [38]
36	Foreground Inpainting†	LaMa [35]
38	Video Generation	LTX-2 [23]
39	Video Object Extraction	SAM 2 [32]

3.3 System Architecture

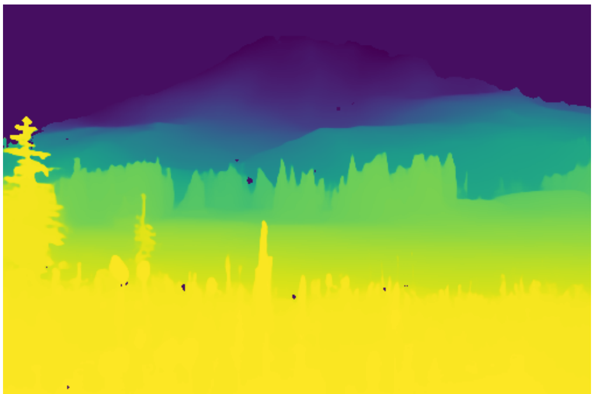
The system splits into a Python generation server and a rendering client. The server runs the pipeline and exposes the result as a `.frame` archive over a WebSocket control channel and an HTTP asset channel, so the client streams assets as they become available rather than waiting for the whole scene. A Unity client (C#, URP) targets desktop VR and carries the runtime shading: a terrain splat shader blending the layered material of section 6, an animated water surface, and wind-driven foliage sway on one global wind field. It also runs the scene's physics: the terrain mesh carries a collision surface, so placed objects rest on the ground rather than floating above it, and the rigid bodies of section 9 resolve against the terrain and against each other.

4 The Panoramic Data

Every track in fig. 1 consumes the same three artifacts: a 360° panorama, an aligned metric depth map, and a per-pixel semantic typing. The depth map and semantic typing are both derived from the generated panorama, which extends the provided input image into a complete 360° view of the scene.

We use one capture, a photograph of Mount Rainier, as a running example throughout. Figure 2 shows the input and the panorama generated from it: the photograph covers roughly a sixth of the sphere, and everything else is synthesized.

(a) Input photograph.



(b) Perspective depth, the metric anchor.

Figure 2: The Mount Rainier example. The input photograph (a) and its perspective depth estimate (b) are outpainted into a full equirectangular panorama (below). The photograph occupies the region around the mountain; the remaining ∼83 % of the sphere is generated.

4.1 Panorama Generation and Metric Calibration

The input image is first processed by generating a synthetic caption and a perspective depth estimate, then outpainting the image to a full equirectangular panorama, which is super-sampled to its working resolution. Panoramic depth is estimated on the result [25].

Depth Any Panoramas is built for equirectangular imagery, but its output is relative rather than metric. We calibrate it against Depth Anything 3 [24] run on the original input, which gives a metric baseline over the region the photograph actually observed. The two are related by a monotonic response curve rather than a single scale factor, because the relationship is not linear.

Fitting the response curve. Let $r \in [0, 1]$ be the panoramic model's raw prediction and d the metric depth from the perspective model. We project the hero photograph into the panorama using the same field of view that was used to place it during outpainting, which gives a set of corresponding samples $\{(r_i, d_i)\}$ over the overlap region. Sky pixels and non-finite or non-positive samples are discarded. The surviving samples are partitioned into $N = 14$ quantile bins of r, and each bin contributes one knot

$$c_b = \text{median}\{r_i : i \in \mathcal{B}_b\}, \qquad m_b = \text{median}\{d_i : i \in \mathcal{B}_b\}, \tag{1}$$

where $\mathcal{B}_b$ is the set of samples in bin b. Medians rather than means keep mismatched samples, typically foreground objects surviving into the overlap, from dragging a knot, and a running maximum $\tilde{m}_b = \max_{k \le b} m_k$ enforces that depth cannot decrease as the raw prediction increases. Calibration is attempted only above 300 valid samples; below that the overlap is too small to fit reliably and the depth is left uncalibrated. Metric depth is then recovered by interpolating $(c_b, \tilde{m}_b)$. The curve is treated as a property of the model rather than of the image, so the same fit applies to every panoramic depth map in the scene, which is what keeps them on a common scale.

Extrapolating beyond the fitted range. The overlap region only ever covers the photograph's own near field, a few to a few tens of meters, so it never contains genuinely distant terrain. Interpolation alone would flat-clamp every raw value past the fitted range onto the curve's last knot, collapsing a mountain a kilometer away and a tree just past the fitted range onto the same depth. We instead extrapolate in log-depth space, since the raw prediction approaches its ceiling asymptotically as true depth grows,

$$d(r) = \exp\big(\log \tilde{m}_N + s\,(r - c_N)\big), \qquad s = \frac{\log \tilde{m}_N - \log \tilde{m}_{N-k}}{c_N - c_{N-k}}, \tag{2}$$

with the slope s estimated from the last $k = 3$ knots, and symmetrically at the near end, capped at $d \le \lambda\,\tilde{m}_N$ with $\lambda = 20$. The cap matters because the raw output saturates well before true distance does: a mountainside a few hundred meters out and a hard-clamped sky pixel read at essentially the same ceiling. A slope estimated from a few tail bins and then exponentiated over that gap is extremely noise-sensitive, and produced fabricated depths of tens of kilometers for every pixel sharing a saturated value. One outlier of that size dominates anything treating panorama depth as a trustworthy scale, silently discarding real terrain as out of bounds rather than merely far. The cap recovers no distance the model cannot resolve, but it keeps far content bounded and ordered.

Fitting the scene into the terrain budget. The calibrated map must finally fit inside the terrain grid's footprint, $D_{\max} = 100\,\text{m}$, half the $200\,\text{m}$ grid extent. A uniform rescale preserves relative distances exactly and is the wrong invariant to protect: the scale would be set by the most distant thing in the scene, effectively unbounded for a landscape, and it invalidates the pipeline's second and more load-bearing anchor, the assumed camera height that places ground at $-h$ below the camera. On Rainier, fitting a summit kilometers away into $100\,\text{m}$ divided the map by $6.5\times$: ground $1.37\,\text{m}$ below the camera returned $0.21\,\text{m}$ of depth, putting the reconstructed surface $0.9\,\text{m}$ *above* eye level with the camera buried in its own terrain. A flat clamp fails oppositely, collapsing a $120\,\text{m}$ ridge and a $900\,\text{m}$ peak onto one wall.

We compress only the far end, leaving the near field metrically intact. With a knee at $k = 0.6\,D_{\max}$ and tail $t = D_{\max} - k$,

$$d' = \begin{cases} d, & d \le k, \\ k + t\left(1 - \exp\left(-\dfrac{d-k}{t}\right)\right), & d > k. \end{cases} \tag{3}$$

This is C^1 at the knee, strictly monotonic, and asymptotic to $D_{\max}$, so no pixel exceeds the budget without a hard clamp. Distant content keeps its ordering; it simply stops being proportional, which it never meaningfully was once a mountain kilometers away had to fit a 200 m grid. Because the curve is fixed by $(D_{\max}, k)$ rather than by the data's maximum, saturated sky cannot contaminate it, and applying the same function to both depth maps places the same point at the same distance in each.

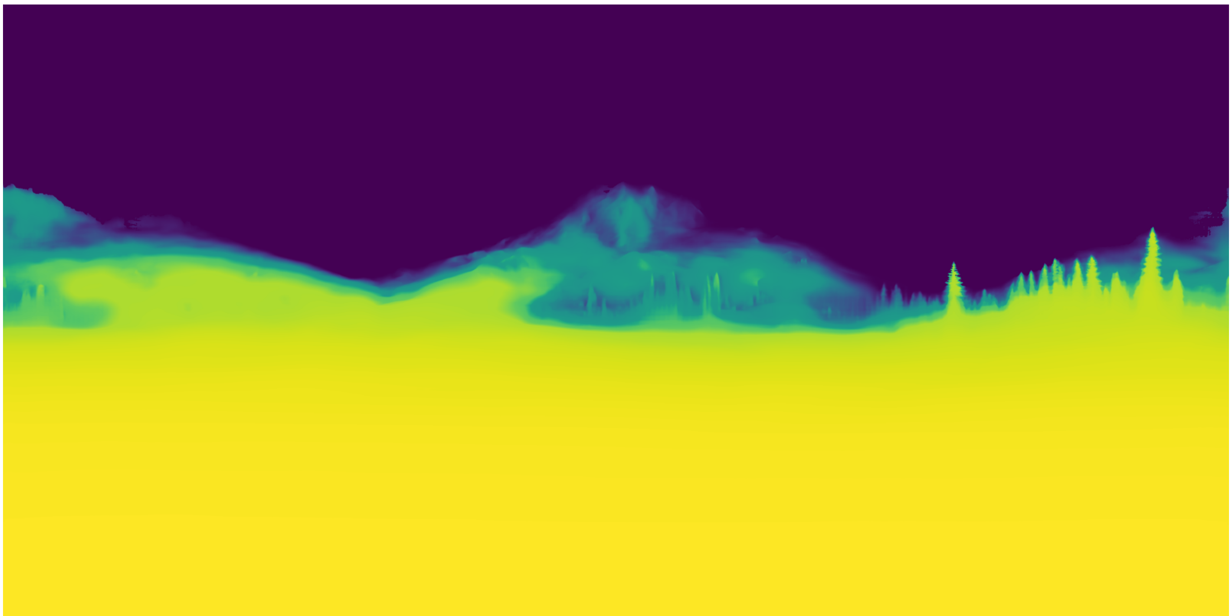


Figure 3: Calibrated panoramic depth for the Rainier capture, near field bright. Two properties discussed above are visible directly. The near-field meadow is resolved in detail, while everything from the mid-ground treeline outward collapses into a narrow band of values: the raw prediction has saturated, and the far-range compression maps that whole range onto the approach to $D_{\max}$. The sky is masked out rather than assigned a depth.

Consequences. The result is metrically compressed: near distances are accurate, far ones systematically shrunk. Since the user explores only a few meters of the scene the visual effect is minimal, relying on forced perspective, but it is not free. Section 10 returns to a defect it causes, where object size is computed as angular extent times depth and inherits the same shrinkage.

4.2 Object-Removed Panorama

To assist terrain generation we produce a second panorama with foreground objects removed, using ObjectClear [45], which removes objects together with their effects such as reflections and cast shadows. Objects are selected by depth threshold, taking those closest to the camera.

Depth taken from the panorama with objects still in it produces spurious values and raised bumps wherever an object stands in front of the ground, since the surface behind it was never observed; removing them first gives the depth model an unobstructed view. Figure 4 shows the effect: the near-field meadow dominating the lower half is replaced by the bare ground the terrain track needs. The edit is destructive by design. The flowers are not discarded but reconstructed separately as a ground-cover population (section 8), and the region typing that places them is read from the *original* panorama.

4.3 Semantic Region Map

SegFormer [40] assigns per-pixel region types (sky, water, terrain, ground, vegetation, built objects, road, trail) over the panorama. This map drives much of the downstream scene understanding, from terrain construction to object placement.

5 Terrain Geometry

5.1 Projection with UV Retention

The height field is a 4096^2 top-down grid spanning 200 m. Each panorama pixel below the horizon is unprojected to a 3D point using calibrated depth and accumulated into the grid cell containing

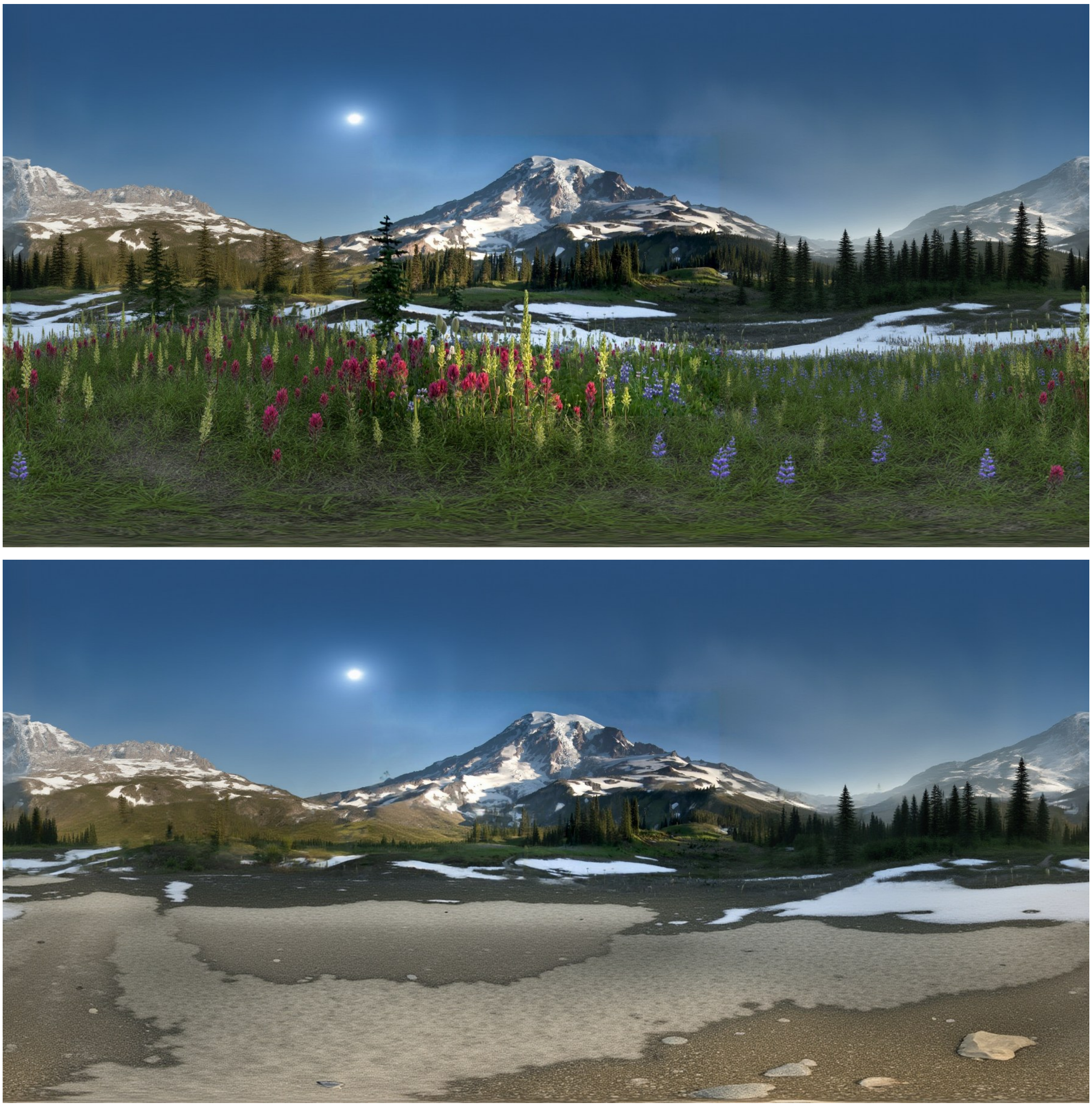


Figure 4: Foreground removal. The original panorama (top) and the object-removed panorama (bottom) used for terrain reconstruction. The near-field meadow and the individual trees have been removed together with their shadows, exposing the ground surface behind them. Depth taken from the top image would place the flower canopy, not the ground, at the bottom of the height field.

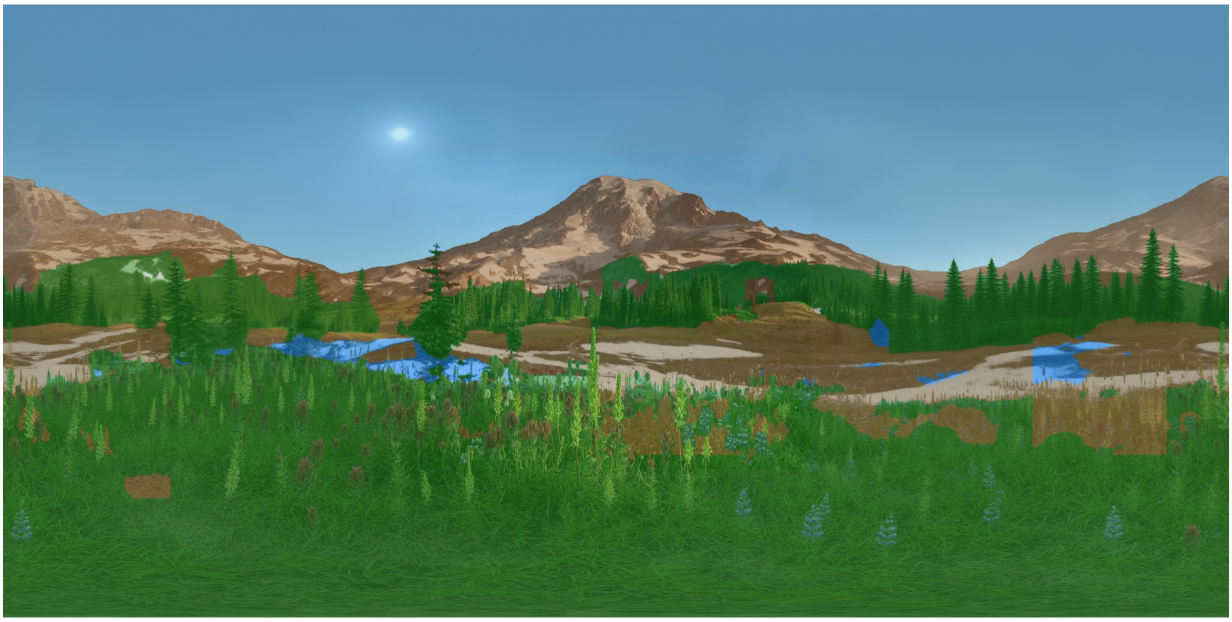


Figure 5: Resolved semantic region typing for the Rainier capture, overlaid on the panorama. Vegetation, bare terrain, snow and standing water separate cleanly in the near field. The typing is run twice, once on the original panorama and once on the object-removed one, because the two answer different questions: where ground cover should be scattered is a property of the scene as photographed, while what the terrain surface is made of is a property of the scene with the objects taken away.

its (x, z). Pixels typed as sky are excluded, since they have no ground intersection to unproject onto.

For every cell that receives a genuine measurement we store the panorama (u, v) it was observed through, together with a mask distinguishing directly measured cells from those later filled by interpolation. This correspondence is what the terrain material of section 6 samples.

5.2 Constrained Harmonic Completion

The projected height field is sparse and uneven: dense near the camera, sparse at range, and empty wherever an object occluded the ground. Completion is posed as a constrained boundary-value problem. Cells with direct, high-certainty measurements become Dirichlet boundary conditions; distant ridge anchors extracted from the sky silhouette are added as further fixed values; the remainder is solved as a discrete harmonic problem

$$\nabla^2 h = 0 \quad \text{on } \Omega_{\text{unknown}}, \qquad h = h_{\text{obs}} \quad \text{on } \partial\Omega_{\text{obs}}, \tag{4}$$

using a sparse linear solve. Measured terrain is therefore never modified, and unobserved terrain is the smoothest surface consistent with it.

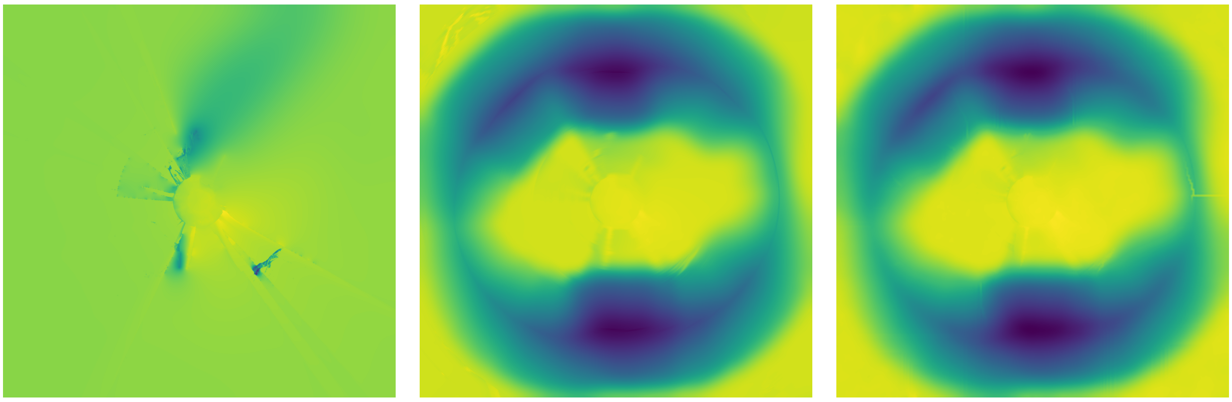
5.3 Ridge Anchoring and Its Assumption

Terrain beyond depth range has no usable measurement, so its elevation comes from the sky boundary: for each panorama column, the first non-sky pixel below sky gives an elevation angle, placed at a fixed radius on the grid. This is a deliberate scale compromise: a mountain kilometers away cannot be walkable terrain in a 200 m grid, so it is brought close and shrunk, preserving silhouette shape rather than true distance. We work around the limitation rather than solve it: past a few hundred meters the depth model has saturated and no longer provides usable values, so the silhouette is the only signal left.

The assumption embedded here is that the sky boundary is the crest of *ground*.

5.4 De-spoking

The equirectangular-to-grid projection maps each panorama column onto a radial line on the grid. Any per-column bias in the depth prediction therefore becomes a radial streak, and because those



(a) Projected observations. (b) After harmonic completion. (c) After erosion refinement.

Figure 6: Height field completion for the Rainier capture, viewed top-down over the 200 m grid with the camera at the center. Direct measurements cover only 7 % of the grid, concentrated around the camera (a); the radial structure there is the projection of panorama columns onto the grid, and the streaks are the spokes removed in section 5.4. Completion supplies the rest (b), and erosion adds drainage structure to the invented relief (c). The difference between (b) and (c) is deliberately small: erosion is restricted to unobserved cells and modulated by the ridge silhouette, so it perturbs the completed surface rather than reshaping it.

streaks lie in the observed data (around 89% of solve nodes are fixed), the harmonic solve cannot remove them.

We remove them with a targeted filter that subtracts only the tangential high-frequency component, computed as a polar round-trip difference

$$h' = h - \left(P(h) - G^{\theta}_{\sigma}(P(h))\right), \tag{5}$$

where P is the polar resample and G^{θ}_{σ} an angular Gaussian blur. Taking the difference cancels the resampling error, which would otherwise bake in concentric rings: resampling error is around 1.2 m against a spoke signal of roughly 0.2 m. Validated end-to-end, the radial elevation profile is unchanged (18.1 m/30.7 m/21.8 m at $r = 45/60/80$ m against 18.1 m/30.7 m/21.9 m before), the cliff mask is bit-identical, and the mean absolute change is 0.19 m concentrated in the spoke regions.

5.5 Erosion Refinement

Harmonic completion is smooth by construction, which is correct as an estimator and wrong as terrain. Real landforms carry drainage structure. We therefore run fluvial incision over the completed field using Landlab [2, 8], restricted to unobserved cells so measured ground is preserved. Erosion strength is modulated by a jaggedness map derived from the ridge silhouette, so the invented relief matches the character of the visible terrain.

5.6 Meshing

The height field is triangulated by Poisson-disc sampling with a near-camera density bias, then Delaunay triangulation, giving natural level-of-detail without axis-aligned artifacts. Water regions are extracted as a separate surface, and the terrain beneath is depressed so an animated water plane never clips through its bed. Disconnected landmasses above a plausibility threshold are extracted as separate formation meshes.

Each vertex carries a second UV channel holding the retained panorama coordinate from section 5.1. Because that channel is per-vertex and interpolated across triangles, triangles spanning an occlusion boundary, where grid-adjacent vertices observed very different panorama rows, smear a wide band of image across a small piece of ground. We detect these geometrically, comparing each triangle's UV span against the span its own size and distance predict, and collapse the offending corner. Detection must be geometric rather than a fixed pixel budget, because a near triangle legitimately spans tens of degrees while the same triangle at range spans under two.

6 Terrain Appearance

6.1 Why Projection Alone Fails

The obvious way to texture the ground is to project the panorama onto it. This fails in two distinct ways.

First, *grazing incidence*. Equirectangular rows compress with distance: at 80 m a single panorama row covers roughly 12 m of ground when the terrain is flat, against 0.5 m when the horizon is anchored at its true elevation. The result is radial smearing.

Second, and less obviously, *upright content*. Projecting assumes the image at elevation φ is ground at radius $h/\tan(-\varphi)$. That holds for bare terrain and fails completely for anything vertical: a 0.5 m flower spans a range of elevations, each landing at a different ground radius, so one flower smears into a streak meters long. Rendered top-down this produces a radial pinwheel.

6.2 Layered Splat Material

Texturing is therefore a weighted blend of layers. A panorama layer supplies real photographic color where projection is trustworthy, gated by a visibility weight that falls off near the nadir and the horizon and is withheld where a texel's own panorama pixel is vegetation or built structure rather than ground. Material layers, one per region type present in the scene, supply the rest.

Those layers are filled by preference rather than by a single method. Real material is preferred wherever the scene provides it (section 6.3); a generated tile is the fallback for a region type with no usable real reference, which in practice means a type that appears only in the far field or only behind an occluder.

Generation is a two-pass process built on FLUX [5] and grounded in the photograph rather than in the region label alone. A handful of real crops of the region type are cut from the panorama and tiled edge to edge across the tile's interior, so the first pass has only to outpaint a thin border ring rather than extrapolate an entire tile from one small inset. The first crop is also captioned with Florence-2 [39], which replaces the generic per-region phrase in the prompt with a concrete description of the material actually present, so a region typed merely as *ground* is generated as, say, cracked sun-bleached mud with small pebbles. The second pass then repaints the whole tile, including the original crop regions, to drive it toward genuine tileability and a uniform finish.

The prompt fixes the framing as an overhead flat lay under diffuse light, with no horizon, no sky and no directional shadows. This matters for the same reason de-lighting matters in section 9: the tile is used as an albedo that the client will light, so any shading baked into it would be applied twice. Where a tile is generated it repeats across the terrain at a fixed frequency, unlike the real-material layer below, which is a single world-space bake.

6.3 Real-Material Pattern Texturing

Where the panorama layer is withheld, we prefer real material over generated material. Following pattern-based texturing [28], small reference patches are extracted from the most solid interior of each region in the panorama and flattened, then assembled over the mesh's own triangles with blended borders so the tiling does not read as faceted.

Reference patches are drawn from the object-removed panorama with real ground color restored from the original wherever foreground removal ate the ground rather than an occluder. Where an occluder genuinely was removed, its footprint is refilled by inpainting from the surrounding real ground.

7 Object Decomposition

7.1 Detection, Typing and Verification

Objects are segmented on the original panorama with SAM 2 [32], tiled and merged to handle equirectangular distortion. Each crop is classified by CLIP against a fixed category set, with a BLIP caption [20] as fallback where the image pass is indeterminate. Labels are then verified by asking Grounding DINO [26] to localize the proposed label in the crop: CLIP proposes, an independent model disposes.

The reason for verification is that CLIP scores a fixed label set and always returns a ranking, so even where it has no real signal some label still wins. The ranking alone cannot distinguish a confident match from a near-uniform one, which is why the decision is referred to a model that has to find the label in the image rather than merely rank it.

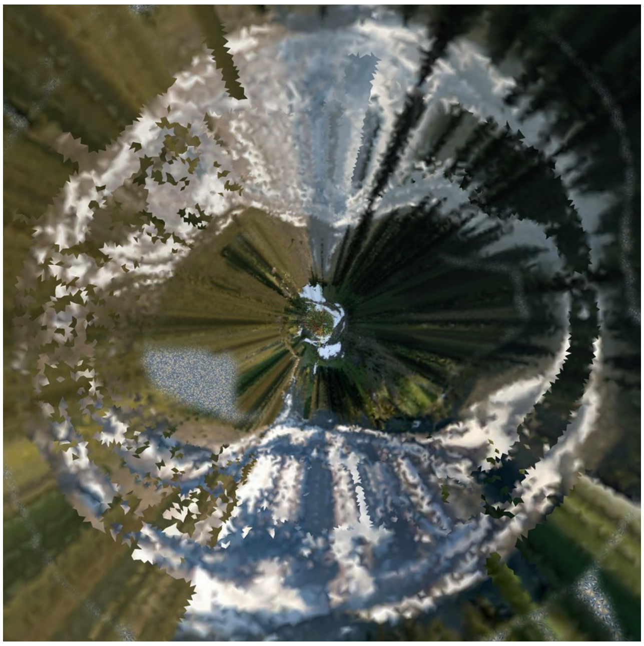


Figure 7: The baked terrain layer for the Rainier capture, viewed top-down in texture space with the camera at the center. The radial structure is the signature of the equirectangular-to-ground mapping: each panorama column becomes one ray on the ground, so upright content along that column is drawn out along it. This is the artifact the layered material and the pattern-texturing path are meant to suppress; section 10 quantifies how much of the terrain is still affected.

7.2 Correlation and Clustering

Surviving detections are grouped by class, then sub-clustered by visual similarity using DINOv2 embeddings and agglomerative clustering, splitting a class into appearance variants (flower colors, tree species). Bucket counts are bounded, and undersized buckets folded into their nearest neighbor, because an unbounded distance cut produces one bucket per instance, measured at 302 groups on one capture, each demanding its own reconstruction.

7.3 Asset Generation and Level of Detail

One mesh is reconstructed per (class, bucket) group with SAM 3D [36] and instanced at every member's position. Two selection decisions matter.

The crop fed to reconstruction is chosen by *resolution* among those passing quality gates, not by classifier score, because a single-view reconstructor given too few pixels does not return a worse object but a different one: a flat sheet. And the representation decision is made on *projected angular size* rather than distance, since distance only proxies resolvability when every object is the same size.

That second decision picks one of three representations, in order of preference. An instance is placed as a reconstructed *mesh* when its group has one and it renders large enough to resolve as geometry. Failing that it takes a *crossed card*: three fixed planes carrying the group's own photograph, twelve triangles that hold their silhouette from any angle. A *billboard*, a single quad that rotates to face the viewer, is the last resort. Keeping the cheap tiers is what makes the geometry budget hold without deciding resolvability by a proxy.

Cards are published only for vegetation, because vegetation is where both other options fail. Single-view reconstruction of a conifer returns a sheet, under a tenth as thick as it is wide on the Rainier capture, which the shape gates correctly reject; and in stereo at the distance trees actually stand, a camera-facing quad reads as flat and swings as the head moves. A rejected reconstruction therefore falls back to a card rather than a billboard whenever its class has one, which is why the same rejection yields cards for Rainier's trees and billboards for Shark Fin's rock fragments (table 3). Because a card is an ordinary textured mesh, it also inherits the sway rig of section 9 instead of standing still beside animated neighbors.

Figure 8: Detections on the Rainier panorama after appearance clustering, drawn in panorama pixel space and colored by bucket. Segmentation runs on the original panorama rather than the hero photograph, so detections cover the full 360°. Distinct buckets separate the conifers on the skyline from the flowering plants underfoot, and split the flowers themselves by color; each surviving bucket costs one reconstruction. The density in the near-field meadow is what motivates the population synthesis of section 8 rather than per-instance reconstruction.

8 Population Synthesis

A detector finds tens of flowers in a meadow that contains thousands. Placing only what was detected leaves a scene correct everywhere it was measured and empty everywhere it was not, which reads worse than either extreme: the eye sees the sampling pattern rather than the meadow.

Two mechanisms fill that gap, and they are deliberately separate because they answer different questions. Population synthesis asks *where else would instances of this class be, given where these ones are*. Ground cover asks *what is the ground made of*, a question with no discrete instances to generalize from at all.

8.1 Synthesis from Observed Statistics

For each (class, region-type) pair with enough real detections to constitute evidence, we measure the spatial statistics of those detections as a pair-correlation histogram, and synthesize additional instances matching them over the rest of the matching region. This is a point-process formulation [7], closely related to appearance-guided element arrangement [14] and in the spirit of example-based world synthesis [4].

8.2 Ground Cover

Grass is not synthesized from detections, because a detector does not usefully enumerate blades of grass. It is scattered by density over a mask, and the interesting part is how that mask is derived.

The mask comes from the region typing of the *original* panorama, for the reason given in section 4: the object-removed panorama has had the meadow erased along with the objects standing in it. But region type alone is too blunt in a way that matters here. The semantic label set folds `grass`, `earth` and `field` together with `sand`, `snow` and `ice` into one ground class, so a snowfield and a meadow are indistinguishable by type, and an alpine meadow is typically ringed by melting snowpack that types identically to the wildflowers beside it.

Color settles it. Each candidate cell is tested against the excess-green of the panorama pixel it was observed through, available directly, because that pixel is the one retained in section 5.1. The threshold is deliberately a low bar rather than a test for lushness: dry grass sits near zero on this scale while sand runs about -0.12 and snow lower still, so it removes only surfaces that are definitively not vegetation. The same measurement is then reused as a *weight* rather than a

(a) Synthesized tree population.
(b) Ground-cover mask.

Figure 9: The two mechanisms on the Rainier capture, both top-down over the 200 m grid. Population synthesis (a) extends the observed tree detections outward across the region where that class is admissible, matching their measured spatial statistics rather than repeating a fixed density. Ground cover (b) is a mask, not a set of instances: it marks where the scattered tuft geometry is admissible. The mask is small relative to the grid because it is derived from the region typing of the *original* panorama, which only sees meadow in the near field the photograph actually observed.

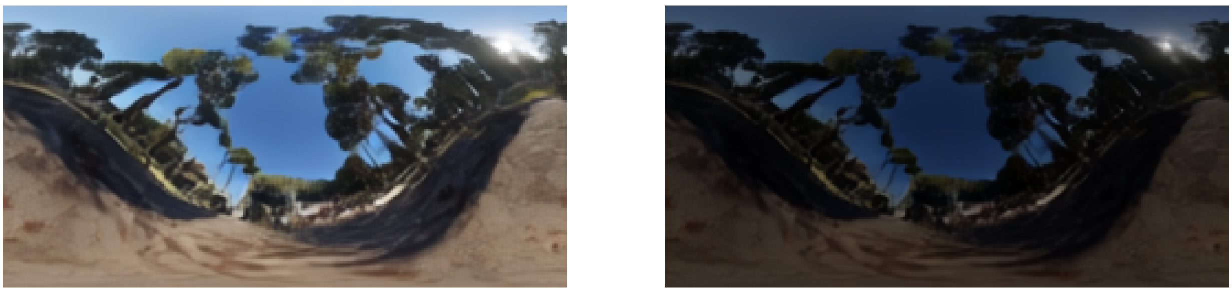
veto, so density tracks how green the ground actually is instead of being uniform over whatever survives the threshold.

Tufts are placed as fixed-orientation crossed cards rather than camera-facing billboards. A billboard is a single quad that turns to face the viewer, which works for an upright subject at eye level and collapses to a line, then swings through the ground plane, for ground cover underfoot.

9 Lighting and Animation

9.1 Illumination and De-lighting

Scene illumination is estimated from the panorama with LuxDiT [22], which recovers an HDR environment and a dominant light direction. The client uses both: the environment as an image-based lighting source, and the direction as a directional light casting the scene's shadows.



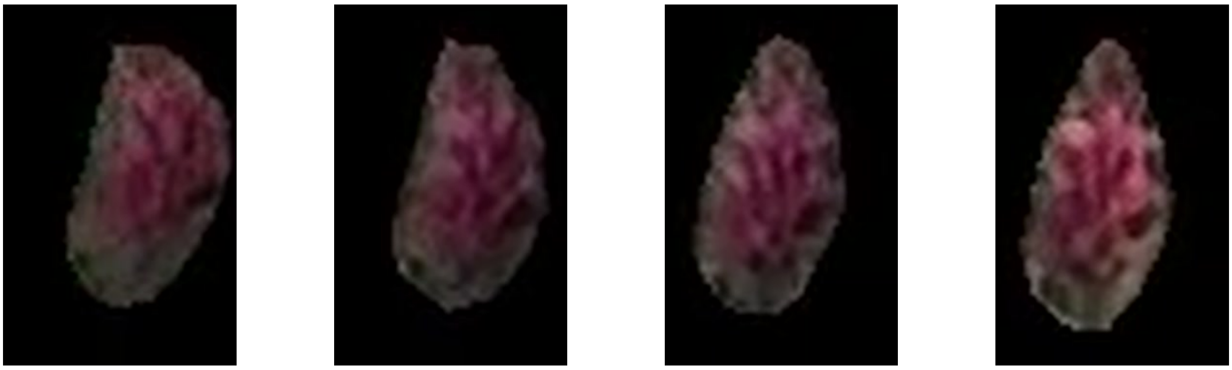
(a) Tone-mapped.
(b) Log scale.

Figure 10: Estimated HDR environment for the Rainier capture. The tone-mapped view shows what the eye would see; the log view exposes the dynamic range the client actually samples, with the sun as a compact high-intensity lobe well above the surrounding sky. The dominant direction extracted from this lobe drives the scene's directional light.

Every asset is cut from a photograph, so it already carries the illumination of the moment it was taken: shadows, directional shading, and the color of the light. Rendering that under a second, freshly estimated light doubles the illumination and reads as flat and over-exposed. Object crops are therefore de-lit with IntrinsicDiffusion [27], so what the client relights is material color rather than a photograph. This is also what makes assets portable: population synthesis places instances anywhere the class is admissible, so an instance rarely sits in the orientation its shading was baked under, and one carrying its original shading would be lit from a different direction than the terrain beneath it.

9.2 Motion

A short video is generated from the panorama with LTX-2 [23], and each detected object is tracked through it with SAM 2 [32]. The camera is held locked off during generation, which is what makes the resulting tracks usable: any drift in an object's image-space centroid is the object moving, not the camera.



(a) 0 s (b) 1.2 s (c) 2.3 s (d) 3.9 s

Figure 11: A single tracked object through the generated video: one paintbrush flower in the Rainier meadow, cropped to the union of its own mask over the whole clip. SAM 2 supplies the per-frame mask, so everything outside the object is removed and what remains is the silhouette the classifier actually measures. The head tilts and recovers between frames while staying in place, which is the signal that decides its type: the centroid oscillates about a stationary mean rather than translating away from it. This flower is therefore typed as anchored and given a sway amplitude and frequency, not a trajectory.

The classification itself is a single measurement. An object whose centroid drifts less than 60 % of its own bounding-box diagonal over the whole clip is treated as *stationary*: it is oscillating in place, anchored to the scene, as a flower or a tree is. Anything that drifts further is a *moving* body.

Stationary objects contribute an amplitude and a frequency read from their own centroid signal, and are rigged rather than animated directly. Each shared bucket mesh gets a three-bone vertical skeleton with joints at its base and at one and two thirds of its height. The top joint sits at two thirds deliberately: a joint bends what is above it, so one at the very top has no geometry to move and degenerates into a dead bone, hinging the upper half about a point instead of curving. With joints at thirds the bottom stays rigid on the ground and each joint inherits the bend below it plus its own. Timing is per instance while the skeleton stays part of the shared mesh, so a meadow of thousands of plants remains one instanced draw.

Moving objects are rigid bodies. Their track is unprojected into world space, holding depth fixed at the reference frame's estimate since no per-frame depth exists, and fitted with a smoothed curve to recover linear velocity and acceleration. Those are the initial conditions handed to the physics engine, which resolves the object against terrain and other objects. Angular velocity is out of scope: it is not reliably recoverable from a monocular silhouette, and a wrong spin is more noticeable than none.

9.3 A Shared Wind Field

Foliage sway, grass motion and the water surface are driven by one scene-global wind field, an ambient base strength plus discrete gust events, rather than by independent per-object oscillators. That is what gives the scene its connectedness: a gust crosses the meadow and everything in its path responds in turn. Independent oscillators are cheaper and read as obviously synthetic, because nothing in a real landscape moves without its neighbors moving too.

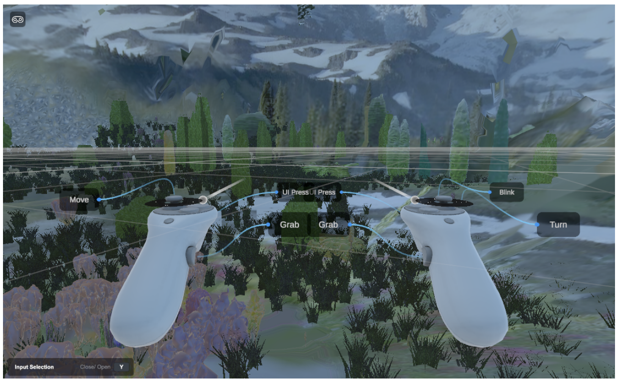
The field is authored rather than derived. The per-object amplitudes above set how strongly each instance responds, but the gust schedule does not yet come from the generated video. Deriving it from the observed correlated motion is left as future work.

10 Results and Discussion

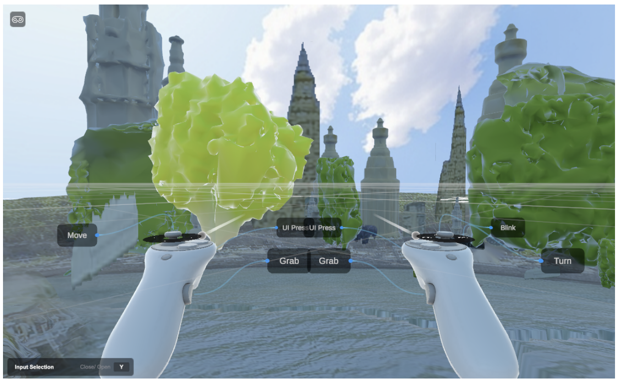
10.1 Evaluation Setup

We evaluate on three captures chosen to stress different assumptions: **Mount Rainier** (alpine meadow, dense vegetation, snow), **Paris** (flat urban river scene, architecture), and **Shark Fin Cove** (coastal cliffs, open water, sea stacks).

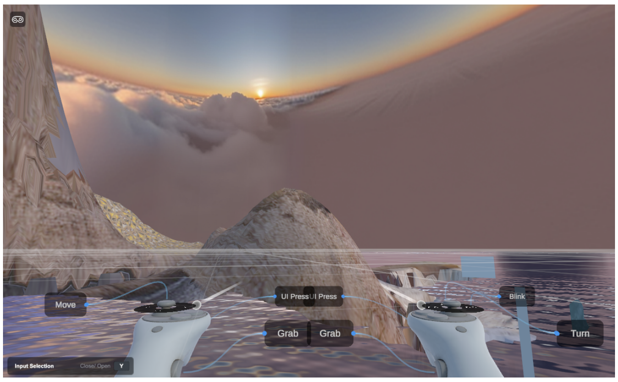
Ground truth 3D does not exist for any of them, so evaluation is necessarily diagnostic: we instrument the pipeline, measure intermediate quantities against what the scene demonstrably contains, and localize defects to specific stages.



(a) Mount Rainier



(b) Paris



(c) Shark Fin Cove

Figure 12: Reconstructed environments viewed from near the capture point.

Figure 12 shows three reconstructed environments viewed from near the capture point.

10.2 Quantitative Summary

Table 3 summarizes scene composition. The mesh/card/billboard split indicates how far object reconstruction got on each capture, with the caveat that the card column is dominated by ground

Table 3: Scene composition per capture. *Relief added* is the elevation range introduced by terrain reconstruction beyond what the height field measured.

Capture	Meshes	Cards	Billboards	Relief added	Dominant limitation
Mount Rainier	128	9 303	33	35 m	Conifer reconstruction
Paris	27	4	56	78 m	Skyline read as landform
Shark Fin Cove	0	1	42	81 m	Detections are rock fragments

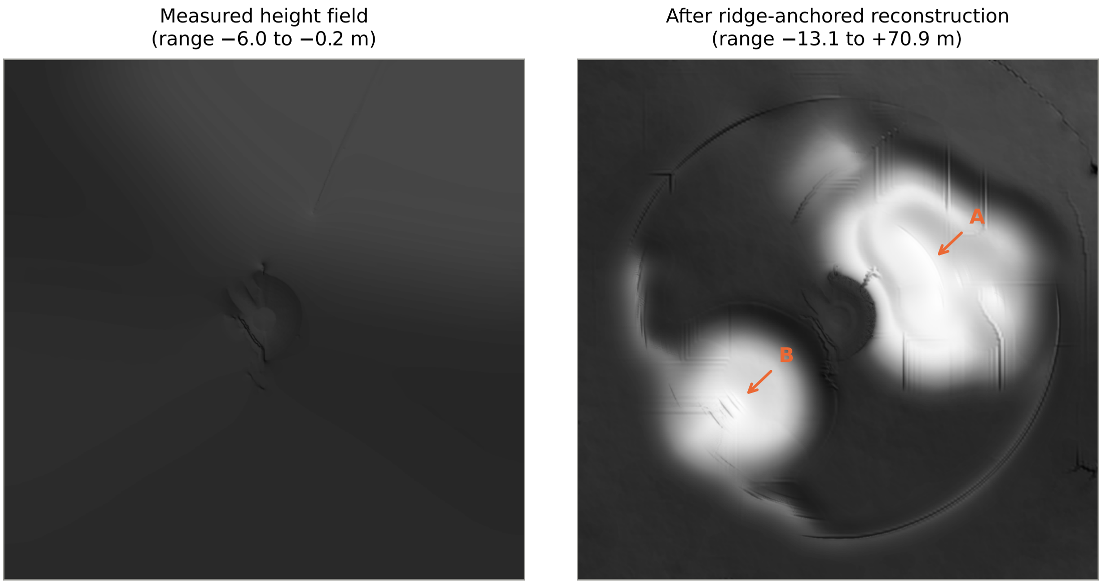


Figure 13: The Paris failure. **Left:** the measured height field, spanning −6.0 m to −0.2 m: correctly flat, and near-featureless under a shared shading scale. **Right:** after ridge-anchored reconstruction, spanning −13.1 m to 70.9 m. Hill A (5966 m^2) is the right-bank treeline; hill B (2674 m^2) is the cathedral. Both panels share one shading scale, so the left panel's flatness is the measurement, not a rendering choice.

cover rather than by object fallbacks.

10.3 What the Measurements Support

Two behaviors are worth stating separately from the corrective measures, which table 4 quantifies in full.

Terrain geometry on landform scenes. Where the horizon genuinely is terrain, reconstruction behaves as intended: measured cells are preserved exactly by construction, the harmonic solve produces a continuous surface, and erosion supplies drainage-consistent relief. Mount Rainier's 35 m of added relief lies beyond depth range, where the silhouette is the only evidence available and 98% of it types as terrain.

Ground-cover masking. Color vetoing excludes snowfields, which type identically to meadow under the semantic label set. With the veto ordered after morphological closing, zero grass instances remain on cells it rejected.

10.4 Failure Modes and Their Causes

The more instructive result is where the system fails, because in every case we localized the failure to an *assumption* rather than a model. A defect can also surface far from its cause: object size compression originates in depth calibration (section 4) and manifests at placement, eight stages later. Section 10.5 reports what each corrective measure achieved.

Skyline read as landform. Ridge anchoring assumes the sky boundary is the crest of ground. In Paris it is a roofline 40 m away. The measured height field spans -6.0 m to -0.2 m (flat, correctly), and reconstruction produced -13.1 m to 70.9 m: a $5966\,\mathrm{m}^2$ hill on the right-bank treeline and a $2674\,\mathrm{m}^2$ hill on the cathedral (fig. 13). The discriminator is what the silhouette is *made of*: the fraction of silhouette columns typing terrain is 0% for Paris against 98% for Rainier and 100% for Shark Fin. Paris is not merely lowest; not one column of its horizon is a landform.

Confident misclassification on texture. Shark Fin Cove produced 49 detections at median confidence 0.95; the two plausibility vetoes of table 4 later remove 28 of them, and every instance that survived failed the reconstruction shape gates. The captions are diagnostic: a rock typed *person* is captioned "a piece of pizza with bacon on it"; another typed *aircraft* is "a giraffe that is flying in the sky". On captures with weak image signal the class is derived from the caption (59% of Shark Fin crops and 91% of Rainier's), and the captioner describes unparseable texture fluently and wrongly. These pass label verification, because a tight cutout fills its own box and an open-vocabulary detector will localize almost any label inside it.

The consequence for reconstruction is direct: SAM 3D given a flat rock fragment returns a flat sheet (normalized extents $[1.00, 0.049, 0.425]$), the shape gates correctly reject it, and every instance falls back to a billboard. Shark Fin's zero meshes are the gates behaving as intended on inputs that are not objects.

The signal that separates them is the region map, because it is the only opinion in the chain that does not look at the crop. A per-crop plausibility veto, a class that cannot occupy the region its box sits in, removes 43% of Shark Fin's detections with no false positives across the other captures.

Thin vegetation reconstruction. Conifers fail single-view reconstruction. Given a good 120×326 crop, SAM 3D returns a mesh with aspect 3.50:1 against detections at 0.12:1, a $28\times$ disagreement, so 70 of 75 instances fall back to crossed cards. A thin conifer offers almost no volume cue from one view. Species-aware approaches [19] are the natural remedy.

Water surface following terrain. Water inherited the terrain height under each of its cells, which is correct only if the ground under a lake is flat. On Shark Fin one $11\,872\,\mathrm{m}^2$ body of sea spanned 73.5 m in elevation, with 21.7% of its vertices more than a meter above the median: the ocean draped up the cliffs.

Far-field texture resolution. Even with folds repaired, 67.8% of Shark Fin's terrain area exceeds 0.5 m per texel against 20.9% for Rainier, because a clifftop camera places most of its surface near the horizon where equirectangular compression is worst. The degradation is entirely a far-field effect: within 25 m the two captures are indistinguishable at 0.02–0.07 m per texel, but by the 50–100 m ring Shark Fin reaches 1.42 m per texel against Rainier's 0.17 m.

10.5 Corrective Measures and Their Measured Effect

Each failure above was addressed by making its implicit premise testable from evidence already computed elsewhere in the pipeline, rather than by tuning a threshold or replacing a model. Table 4 reports the measured effect of each, evaluated by replaying the modified stage over saved pipeline state from all three captures. We report effects on *every* capture, not only the one that motivated the change, because the risk being managed is regression on scenes that were already correct.

Two of these deserve comment because the obvious version of each is measurably wrong.

Caption plausibility requires conjunction. Rejecting a detection because its caption mentions something implausible is appealing and unsafe. On Mount Rainier, 21 flower detections carry captions such as "a close up of a flower in a vase", and 19 of those 21 are real, placed flowers. The captioner identifies the *subject* correctly and confabulates the *setting*. Conversely, rejecting on scene-tag support alone is also unsafe: Rainier's tags name no tree, and it has 67 real ones. Only the conjunction of four conditions (no scene-tag support, implausible caption vocabulary, caption not naming the class, and class not vegetation) produced zero false positives. Notably, the caption–class agreement test is unusable as a veto (its synonym gaps delete real objects) but safe as an *exemption*, because a gap then merely retains a spurious detection rather than removing a real one.

Table 4: Corrective measures, the premise each makes testable, and measured effect across captures. "FP" denotes false positives: correct content wrongly removed.

Measure	Premise made testable	Measured effect
Silhouette composition gate	"The sky boundary is the crest of ground"	Terrain-typed silhouette fraction: Paris 0%; Rainier 98%; Shark Fin 100%. Ridge anchoring suppressed on Paris only; other captures bit-identical.
Per-crop region plausibility	"This class can occupy the region its box sits in"	Removes 9/109 (Paris), 4/237 (Rainier), 12/49 (Shark Fin). 0 FP; Rainier's four are phantom aircraft inside the snowfield.
Caption–scene plausibility	"The caption describes something this scene can contain"	Removes a further 16 (Shark Fin), 9 (Rainier), 2 (Paris). 0 FP. Requires four conjunctive conditions; each weaker variant deletes real content (see text).
Water surface levelling	"The ground beneath water is flat"	Shark Fin elevation spread 73.5 m $\rightarrow$ 0.00 m; resulting shoreline discontinuity 0.14 m. Paris (already near-level) 1.67 m $\rightarrow$ 0.55 m across 12 bodies.
Projected-size LOD	"Distance predicts resolvability"	Rainier meshed instances 545 $\rightarrow$ 296; triangles 4.47 M $\rightarrow$ 2.48 M; per-class override table eliminated.
Resolution-based asset selection	"The best-scoring crop is the best reconstruction input"	Source crop area increased 1.0–9.2$\times$ per group; 205 of 471 instances had been rejected as degenerate reconstructions.
Geometric UV-fold repair	"Adjacent grid cells observed adjacent panorama rows"	Collapses 5624 triangles (Rainier, 7.0%), 2169 (Shark Fin), 1479 (Paris). Converges in two passes; face count unchanged.
Occluder-footprint inpainting	"What foreground removal left behind is ground"	Below-horizon pixels showing fabricated fill: 23.9% $\rightarrow$ 17.1%.

Robust estimation must respect sampling density. Water levelling initially estimated each body's surface from its mesh vertices. Because the mesh is Poisson-disc sampled with a near-camera density bias, 4309 of Shark Fin's 11 366 water vertices lie within 20 m of the camera, and the resulting estimate described the cove at the viewer's feet rather than the sea: -4.54 m against a shoreline at -0.99 m. Estimating over the uniform grid instead gives -1.13 m and a 0.14 m shoreline step. The estimator was correct; the sample weighting was not.

11 Conclusions and Future Work

11.1 The Case for a Hybrid Pipeline

The result we most want to draw out is the shape of the system rather than any single stage. Into Frame mixes learned models and classical graphics roughly two-to-one by stage count, and neither part would produce this output alone.

Generative models make the problem tractable at all. Five sixths of the sphere and the ground behind every object are never observed, and no classical reasoning recovers content absent from the input. Outpainting, foreground removal, single-view reconstruction and video generation supply it.

What they do not supply is structure. A generative system asked for a scene returns something viewable: a panorama, a mesh, a radiance field. It does not return a collidable surface, a population whose density can be sampled, a water plane animated separately from its bed, or a mesh with a skeleton in it. Every property that makes the output walkable rather than watchable comes from the classical side: a harmonic solve that provably never modifies measured ground, fluvial incision, point-process synthesis fitted to observed statistics, pattern texturing, and a three-bone rig driven by a shared wind field. Of the 38 active stages, 12 use no learned model at all, and those are disproportionately the ones producing physical behavior.

The division of labor is the point. Learned components answer *what is there*, including where nothing was observed; classical components decide *how it behaves*, under constraints a generative model cannot honor: measured terrain must survive completion unchanged, water must sit below its bed, a meadow of thousands of plants must remain one instanced draw. Treating those as requirements rather than as things to be learned is what turns a picture of a place into a place.

It also explains our failures. As section 10 sets out, the surviving defects are mostly not model failures but assumption failures, where a step correct for one scene category is applied silently to another. That is the more tractable problem: an assumption can be identified, measured, and gated.

11.2 Future Work

Explicit scene classification. A first-class scene-type estimate (natural landform, urban, coastal, interior) consumed by every stage that currently embeds an implicit assumption would address the largest class of failure we observed. The system is biased toward outdoor natural scenes, and several assumptions do not hold outside them: that the sky boundary is the crest of ground, that a large uniform region is terrain, that vegetation is the dominant content.

Confidence propagation. Stages produce point estimates. Propagating uncertainty would let downstream stages weigh evidence rather than accept it, and would let the system report where it is guessing.

Better single-view reconstruction for thin structures. Vegetation is the dominant content of natural scenes and the weakest part of our object track; species-aware priors [19] are the obvious direction.

Joint refinement. The pipeline is strictly feed-forward. A closing optimization that adjusted depth, geometry and placement jointly against photometric consistency could correct compounded error that no single stage can see.

Perceptual evaluation. Our evaluation is diagnostic: it measures where the system is wrong, not how the result feels to be inside. A user study of presence, scale perception and navigation comfort would test what the system is actually for, and is the clearest gap in our rigor.

Audio and interaction. The scene is something to move through rather than to act on. Spatialized audio generated jointly with the scene [17], and ways for the user to disturb the environment rather than only observe it, would extend interactivity in the direction this framing implies.

11.3 Closing Remarks

Into Frame demonstrates that a single photograph can become an interactive environment, with terrain, objects, ground cover, water and lighting reconstructed as distinct physical components rather than as one representation serving every purpose. That decomposition buys the depth of the result: each component is reconstructed by a method suited to its role and then behaves according to it.

The remaining failures are mostly not failures of any model, but assumptions that held for outdoor natural landscapes and did not extrapolate beyond them. That is an encouraging place to stop: the boundaries are legible and measurable, and addressing them means making implicit commitments explicit rather than waiting for better models.

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